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Euclidean Geometry In Mathematical Olympiads

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Problem 1.51 (IMO 1985/1)


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illogical_21

1034 posts

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#1

A circle has center on the side $\overline{AB}$ of the cyclic quadrilateral $ABCD$. The other three sides are tangent to the circle. Prove that $AD + BC = AB$.

This post has been edited 1 time. Last edited by illogical_21, Mar 18, 2019, 11:21 AM

illogical_21

1034 posts

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#2

Solution

Let $T \in AB$ such that $AD = AT$, then
 $\angle DTO = 180^\circ - \angle DTA = 180^\circ - \frac{180^\circ - \angle DAB}{2} = 180^\circ - \frac{\angle DCB}{2} = 180^\circ - \angle DCO$
 $\implies CDTO$ is cyclic, so $\angle CTB = \angle CDO = \frac{180^\circ - \angle CBT}{2}$
 $\implies BC = BT$, then $AD + BC = AT + BT = AB$, so we are done.

kootrapali

4527 posts

Mar 3, 2020, 6:23 PM • 1 🍌

#3

Solution

Let O be the center of the circle, and let E, F , and G be the feet of the altitudes from O to AD, CD , and BC , respectively. Notice that the circle must intersect segment AB twice because otherwise AD and BC won't be tangent to the circle. WLOG let the radius of the circle be 1. Let $AE = a$ and $BG = b$. Then $AO = \sqrt{a^2 + 1}$ and $BO = \sqrt{b^2 + 1}$. Since $ABCD$ is cyclic, we have $\angle C = 180^\circ - \angle A$ and since $\angle OFC = \angle OGC = 90^\circ$, $OFCG$ is also cyclic, so $\angle FBG = 180^\circ - \angle C = \angle A$. Consider $\triangle BGC$. Since $OG = 1$, we have

$$CG = \tan \frac{\angle A}{2} = \frac{\sin \angle A}{1 + \cos \angle A} = \frac{\frac{1}{\sqrt{a^2+1}}}{1 + \frac{a}{\sqrt{a^2+1}}} = \frac{1}{a + \sqrt{a^2+1}}.$$

Similarly,

$$ED = \frac{1}{b + \sqrt{b^2+1}}.$$

Now,

$$\begin{aligned} AD + BC &= a + \frac{1}{b + \sqrt{b^2+1}} + b + \frac{1}{a + \sqrt{a^2+1}} = \frac{ab + a\sqrt{b^2+1} + 1}{b + \sqrt{b^2+1}} + \frac{ab + b\sqrt{a^2+1} + 1}{a + \sqrt{a^2+1}} = \\ &= \frac{a^2b + ab\sqrt{a^2+1} + a^2\sqrt{b^2+1} + a\sqrt{(a^2+1)(b^2+1)} + a + \sqrt{a^2+1} + ab^2 + ab\sqrt{b^2+1} + b^2\sqrt{a^2+1} + b\sqrt{(a^2+1)(b^2+1)} + b + \sqrt{b^2+1}}{(a + \sqrt{a^2+1})(b + \sqrt{b^2+1})} \\ &= \frac{(\sqrt{a^2+1} + \sqrt{b^2+1})(ab + a\sqrt{b^2+1} + b\sqrt{a^2+1} + \sqrt{(a^2+1)(b^2+1)})}{ab + a\sqrt{b^2+1} + b\sqrt{a^2+1} + \sqrt{(a^2+1)(b^2+1)}} = \sqrt{a^2+1} + \sqrt{b^2+1}. \end{aligned}$$

Also, $AB = AO + BO = \sqrt{a^2 + 1} + \sqrt{b^2 + 1}$, so $AD + BC = AB$.

bever209

1522 posts

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#4

“ illogical_21 wrote:

Solution

why is $DCB/2 = DCO$?

Mathsciencecl...

1241 posts

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#5

“ bever209 wrote:

“ illogical_21 wrote:

Solution

why is $DCB/2 = DCO$?

He's referring to O as the center of the new circle. By tangency this follows.

Williamgolly

3760 posts

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#6

Bruh am i missing something?